

Mobile Development :

1 : Introduction to Mobile Dev + Dart Essentials



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Outline :

- **Mobile App Development World:**
 - Mobile Market : Statistics
 - Business Aspects for Mobile Dev
 - History of Mobile App Dev
 - Types of Mobile Applications
- **Essentials for Dart Programming :**
 - Variables & Data Types
 - Functions
 - Classes

Section 1

Mobile App Development World





Mobile Market : Statistics

- **As a mobile user : are we addicted to our phones ?**
 - We are becoming **deeply addicted** to our phones
 - Globally, the average mobile user spent **4.8 hours** on their mobile device
 - **85%** of time spent on smartphones is spent on using apps. (As opposed to phone calls.)
 - The average smartphone owner uses 9-10 apps per day and 30 apps per month.
 - The number of downloads for mobile apps in 2022, exceeds 255 billion

Mobile Market : Statistics

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 - We are becoming **deeply addicted** to our phones.
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 - The average smartphone owner uses 9-10 apps per month.

Addicted for various reasons :

- Social connection
- Work and Productivity
- Information and News
- Entertainment
- Fear of Missing Out (FOMO)
- Escapism
- Boredom
- ...



Mobile Market : Statistics

- **As a mobile user : How many users ?**
 - The number of smartphone mobile network subscriptions worldwide reached almost **6.4 billion** in 2022, and is forecast to exceed **7.7 billion** by 2028.
 - More than **86%** of people have **smartphones**.
 - China, India, and the United States are the countries with the most number of mobile users.

Mobile Market : Statistics

- As a mobile user :

COUNTRY/MARKET	TOTAL POPULATION	SMARTPHONE PENETRATION	SMARTPHONE USERS ↴
China	1.43B	68.40%	974.69M
India	1.42B	46.50%	659.00M
United States	338.29M	81.60%	276.14M
Indonesia	275.50M	68.10%	187.70M
Brazil	215.31M	66.60%	143.43M
Russia	144.71M	73.60%	106.44M
Japan	123.95M	78.60%	97.44M
Nigeria	218.54M	38.10%	83.34M
Mexico	127.50M	61.50%	78.37M
Pakistan	235.82M	31%	72.99M
Egypt	110.99M	64.20%	71.24M



Mobile Market : Statistics

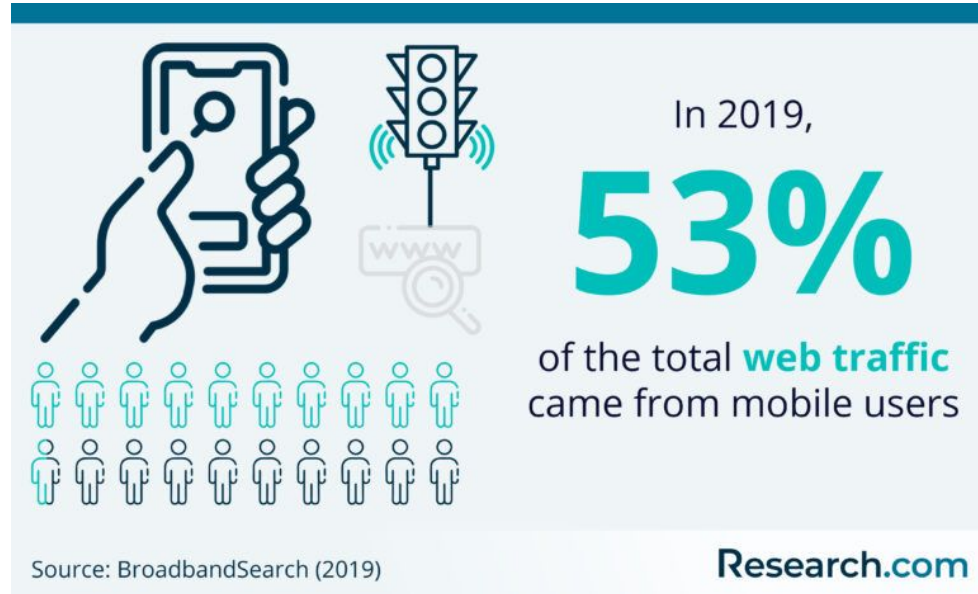
- **As a mobile user : What we do most ? :**
 - 50% of app users mainly use apps for communication (including social networking and chat)
 - There are over 700,000 educational apps on the market.
 - Total hours spent watching video streaming apps grew 32% worldwide.
 - 62% of people have a gaming app installed on their phone.

Stats taken before Youtube shorts, FB reels

Mobile Market : Statistics

- **As a mobile user : Compared to a desktop user:**

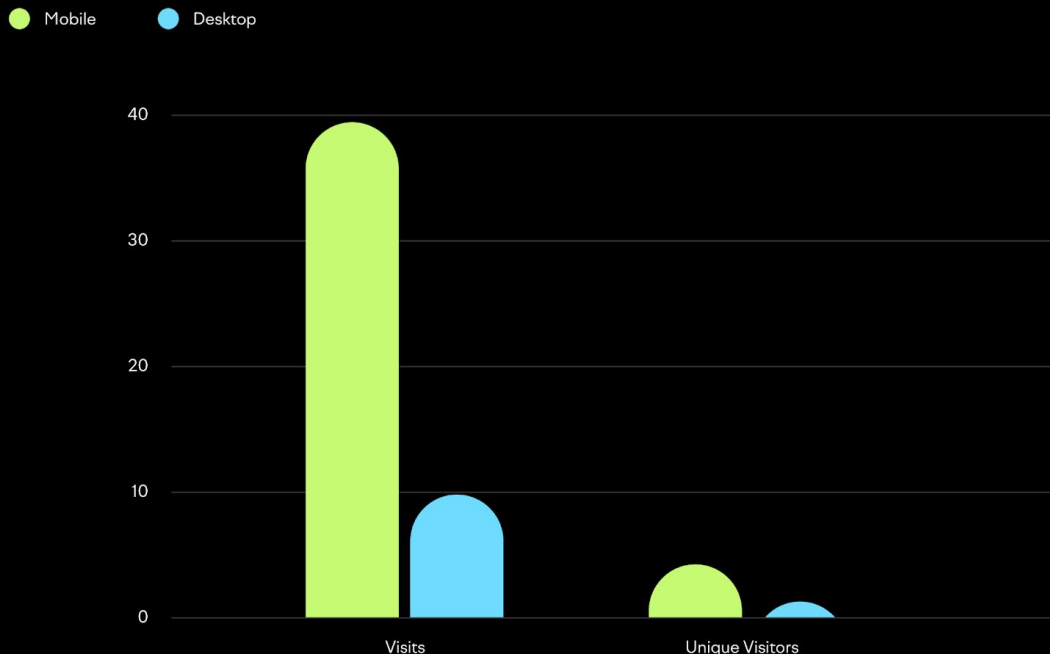
- Significantly more searches are carried out on the mobile phone (**64%**) than on the desktop (35%).
- **53%** of web traffic comes from mobile phones.



Mobile Market

- As a mobile user : Com
 - Significantly more searches are carried out on the mobile phone (**64%**) on the desktop (35%)
 - **53%** of web traffic comes from mobile phones.

Mobile vs Desktop Total Traffic (in billions) (2023)



Source: Semrush Traffic Analytics tool



Mobile Market : Statistics

- **As a company :**
 - Having a mobile app for a company regardless of their activities, is **imperative** for their survival :
 - 33% of American retailers have at least one app,
 - 42% of small businesses now have a mobile app
 - Fortune 500 companies have their mobile apps.
 - Overseas Brands Grow International Footprint, Growing Market Share by 11% in Some Regions



Mobile Market : Statistics

- **As a Mobile App Developer:**

- The number of Mobile Apps is estimated as (As of 24 Sep 2023) :
 - **1.8 million** Apps inside the Apple App Store
 - **3.77 million** apps on the Google Play Store.
- Only **0.5%** of apps are successful*
- App crashes cause 71 percent of uninstalls
- 70 percent of users abandon an app because it takes too long to load (too long = more than 3 seconds)

* <https://www.fyresite.com/how-many-apps-fail/>

Mobile Market : Statistics

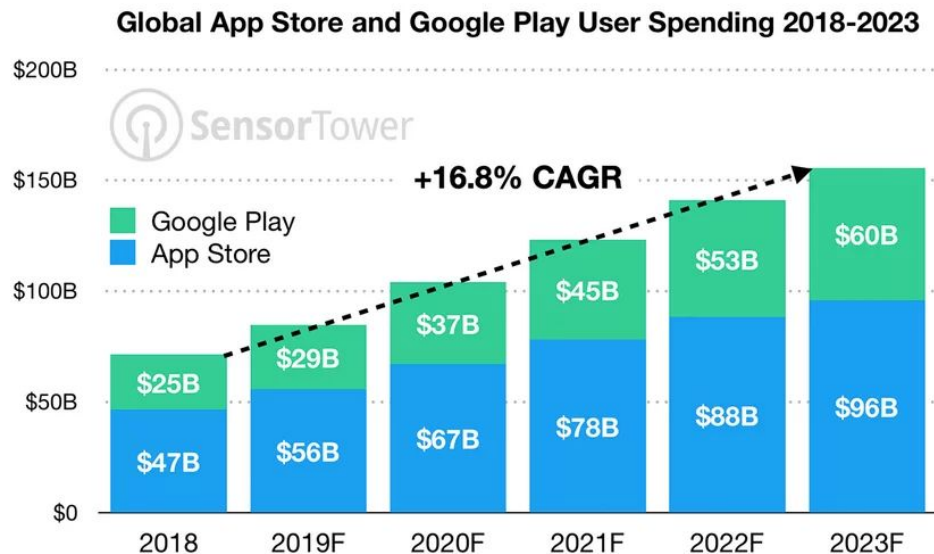
- **As a Mobile App Developer:**

- The market share of mobile OSs, As of January 2023,
 - **Android** holds a **71.63%** market share in the operating system (OS) market.
 - **iOS** has a **27.68%** market share in the OS market.
 - **Samsung**'s market share in the OS market is 0.35%.
 - **KaiOS** holds a 0.11% market share in the OS market.
 - **Windows** has a 0.02% market share in the OS market.
 - **HarmonyOS** is estimated to have 2% in the second quarter of 2023

Business Aspects of Mobile Dev

- How much money in terms of revenue :

- How much user are spending on mobile apps via App Stores.



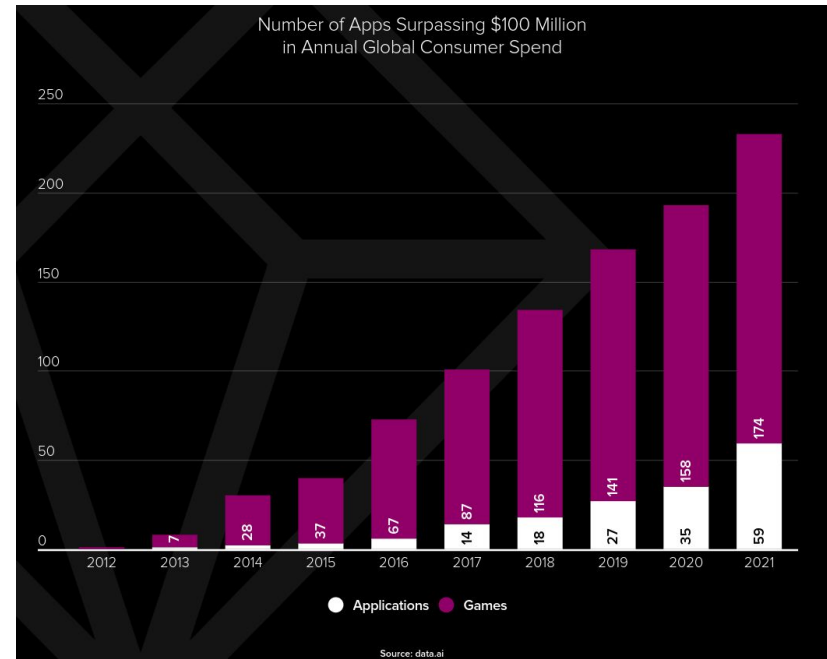
CAGR = Compound Annual Growth Rate

Does not reflect spending on third-party Android stores in China.

Source: Sensor Tower Store Intelligence

Business Aspects of Mobile Dev

- **How much money in terms of revenue :**
 - In 2021, there were over 230 apps and games surpassing \$100M in annual consumer spend, with 13 of them surpassing \$1 Billion.



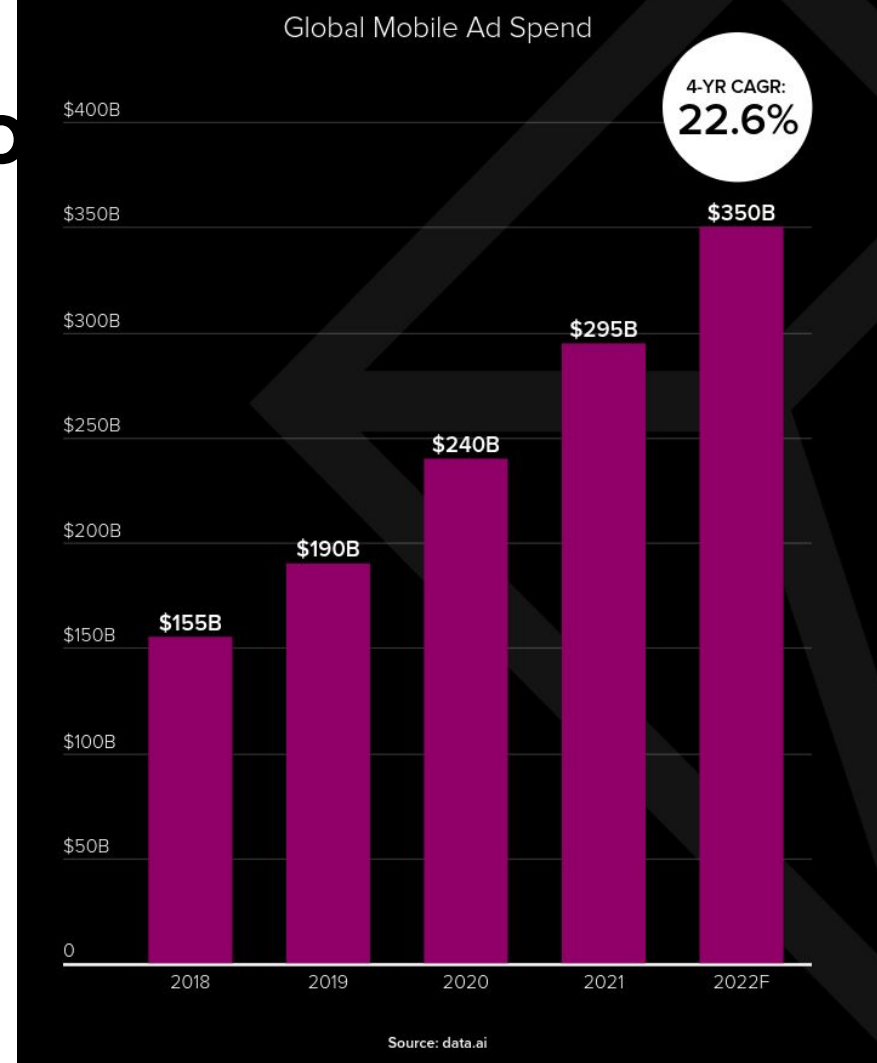
Business Aspects of Mobile Dev

- Startups, Companies and Ventures

App	Revenue (\$bn)
TikTok	2.05
Honor of Kings	1.65
Genshin Impact	1.25
PUBG Mobile	1.12
YouTube	1.05
Candy Crush Saga	1.00

Business Aspects of Mobile Dev

- How much money companies spend in advertising within Apps ?
 - Companies spend over \$350B for advertising in mobile platforms

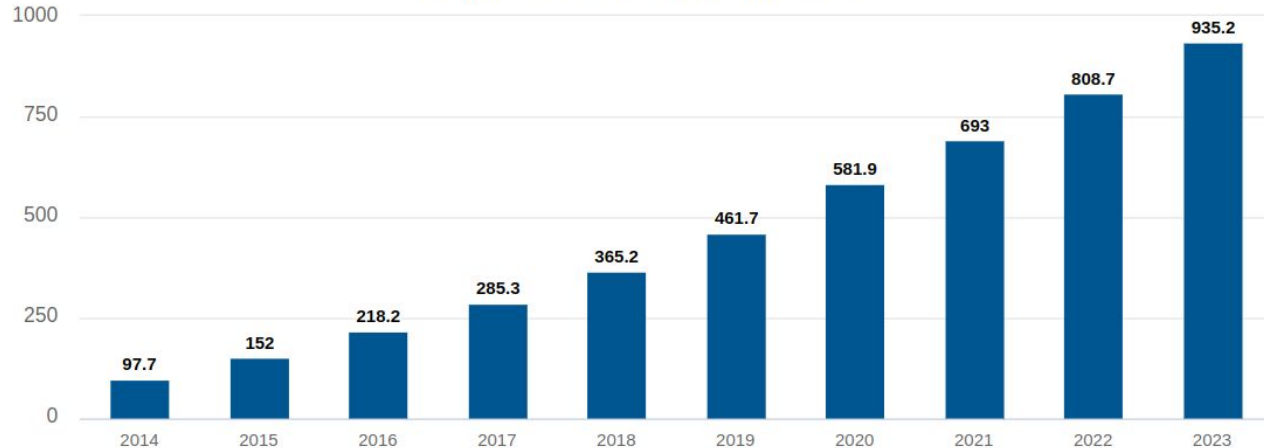


Business Aspects of Mobile Dev

- Other ways of making money ?

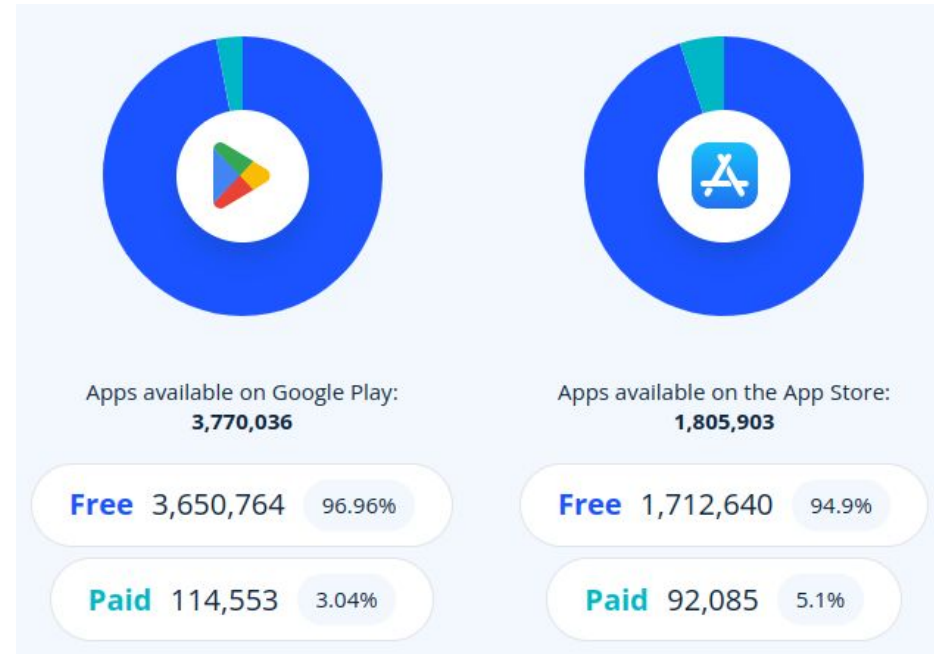
- In-App Purchases
- Subscriptions

Worldwide Mobile App Revenues
From 2014 to 2023, in billion USD



Business Aspects of Mobile Dev

- **Free vs. Paid App ?**
 - Most of the app revenue is coming the free apps.
 - Only a fraction of people are willing to pay for an App.



Business Aspects of Mobile Dev



- Revenue for Development ?
 - Is it cheap to produce a mobile App ?
 - How much money to pay a developer to build an App ?
 - How much mobile app developer gets paid ?
 - How Long does it take to fully develop a mobile app ?
 - Which is better ? Mobile vs. Web Development

Business Aspects of Mobile Dev

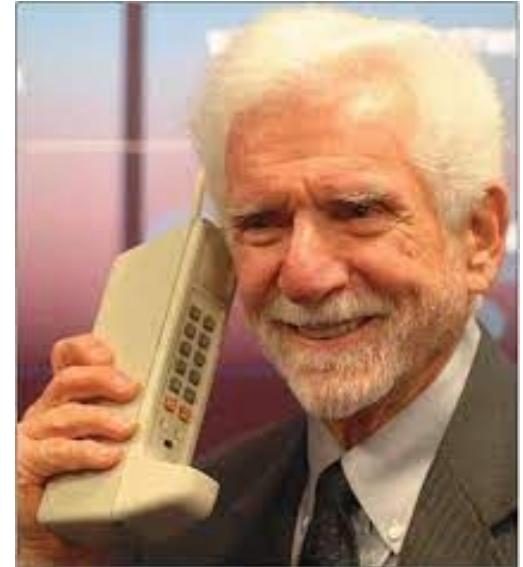
“Mobile is the future of Software Development,”

says Eric Schmidt (Ex-CEO of Google).



History of Mobile App Development

- **1973-1995 : Basic Phone**
 - **1973 : Martin Cooper invented the first mobile phone**
 - 1983 : DumaTAC 8000x is the first commercialized phone.
 - 1993 : IBM announced the first touch phone called IBM Simon with apps : Clock, Calendar, notepad and Calculator, released in 1994



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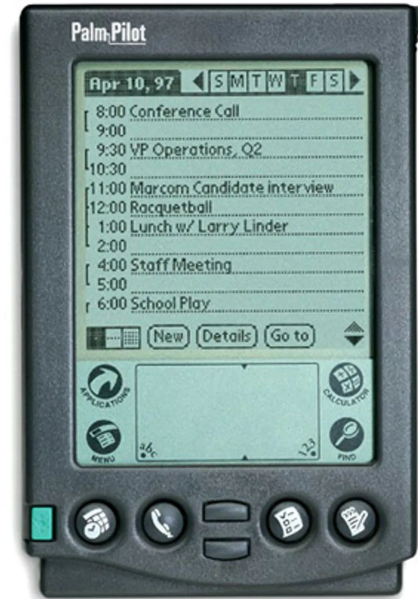
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History of Mobile App Development

- **1996-2006 : Feature Phone**

- **1997 : The PalmPilot was debuted running PalmOS with the ability to run 3rd Party apps.**
- 1999 : Blackberry OS was introduced and made its debut for the Pager Blackberry 850.
- 2000 : Microsoft introduced the tablet PC.
- 2002 : The BlackBerry 5810 smartphone was introduced
- 2003 : Andy Rubin, Rich Miner, Nick Sears, and Chris White launched Android Inc.
- 2005 : Google paid at least \$50 million for Android Inc.



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History of Mobile App Development



- **2007-Now : Smartphone**

- 2007 : Steve Jobs unveiled the iPhone, which had been a long time coming.
- 2008 : Launch of Apple's app store with 552 apps available.
- 2008 : Android is released with its Android Market

Types of Mobile Applications



- **What's a Mobile App ?**
 - A software product designed to install and execute on a mobile device (not necessarily phone) in order to perform certain tasks for the user.

Types of Mobile Applications



- **Apps are meant only for smartphones ?**
 - Mobile Apps are now being developed for various devices including :
 - Tablets
 - Smart watches
 - TVs
 - Cars
 - ...

Types of Mobile Applications



- **Different Types of Apps :**
 - *Native*
 - *Web*
 - *Hybrid*

Types of Mobile Applications



- **Native Mobile App:**
 - Applications which are coded to install and run directly on the operating system without any intermediary layer/app.
 - They are usually coded using a specific programming language and target for a specific platform/device.

Types of Mobile Applications



- **Native Mobile App:**

- Pros :

- *Fast and better performance.*
 - *Easy to benefit from the device/hardware functionalities (Camera, accelerometer, GPS, biometric sensor, Touch/Gesture interactions, Microphone....)*

- Cons:

- *More expensive to develop + Hard to maintain + Extremely difficult to target multiple platforms as you need to maintain different code base.*

Types of Mobile Applications



- **Web App**

- is simply a website that can be accessed from a browser. Web Apps are usually built as responsive as possible to accommodate different screen sizes.
- For Websites, shall we create the desktop/main website mobile friendly/responsive or shall we create a newly dedicated website for mobile users ?
- They need an internet connection to access the website

Types of Mobile Applications

- **Web App:**

- **Pros :**

- Easy, cheap to develop
 - Run on various platforms with a browser
 - Maintenance is easy as it is centralised

- **Cons:**

- No access to the device functionality (or limited now with the advance of web technologies)
 - Cannot work for offline (Though, even it can be done using JS, it is hard to maintain)

Types of Mobile Applications



- **Hybrid or Progressive Web App (PWA) :**
 - is an installable application which makes use of Web Technologies and Frameworks with more rich functionalities, Still, it may requires a backend when needed.
 - It combines the native + web features,
 - Hardware/device functionalities can be accessed.
 - Even though PWA does not need a browser to run, they use a built-in browser component.
 - They can work offline without any internet connection.

Types of Mobile Applications



- **Hybrid or Progressive Web App (PWA) :**
 - **Pros :**
 - Easy to develop and maintain + Cheap
 - Faster build time
 - One code base can be used for various platforms.
 - **Cons :**
 - Performance is questionable.
 - May turn complex depending on the features

Types of Mobile Applications



- **Hybrid or Progressive Web App (PWA) :**
 - **Example :**
 - Gmail
 - Twitter
 - Instagram

A decorative graphic consisting of several colored rectangular bars and a patterned box. At the top left is a pink bar, followed by a blue bar, and then a yellow bar. To the right of these is a black box with a white diagonal line pattern. Below the yellow bar is a blue bar. At the bottom right is a yellow bar.

Section 2

Dart Programming Essentials



Dart Programming Essentials



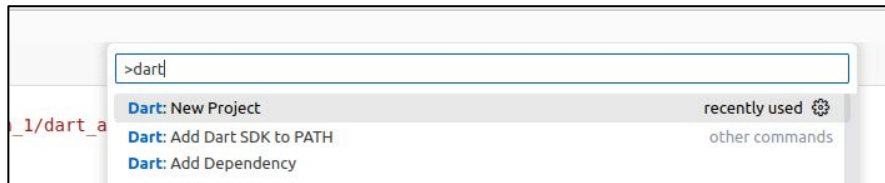
- **Dart as a Programming Language**

- Client-optimized, object-oriented programming for creating apps quickly for many platforms including Android, iOS, Linux, Windows and the web.
- Developed by Google in 2011 to replace JavaScript, but without success.
- Later, It gained huge popularity because it was the main language for Flutter
- Statically typed language where data types of variables must be specified either explicitly or implicitly through inference.
- Supports modern programming features as null safety checks and asynchronous prog.
- Lastly, it is free and open-source + Rich standard library.

Dart Programming Essentials

- **Getting Started with Dart**

- You can start to learn and test dart online at :
 - <https://dartpad.dev/>?
- Install VS + Android SDK + Dart + Flutter on your Machine
- To create new Project, Either :
 - Ctrl + Shift + P and type "dart" → Choose Dart: **New Project**
 - Choose Console App
 - From Console:
 - **dart create project_name**



Dart Programming Essentials

- Getting Started

- Y test

- In

- To

Which Dart template?

Bare-bones Web App web

A web app that uses only core Dart libraries.

CLI Application cli

A command-line application with basic argument parsing.

Console Application console

A command-line application.

Dart Package package

A package containing shared Dart libraries.

Server app server-shelf

A server app using package:shelf.

- From Console:

- **dart create project_name**

>dart

Dart: New Project

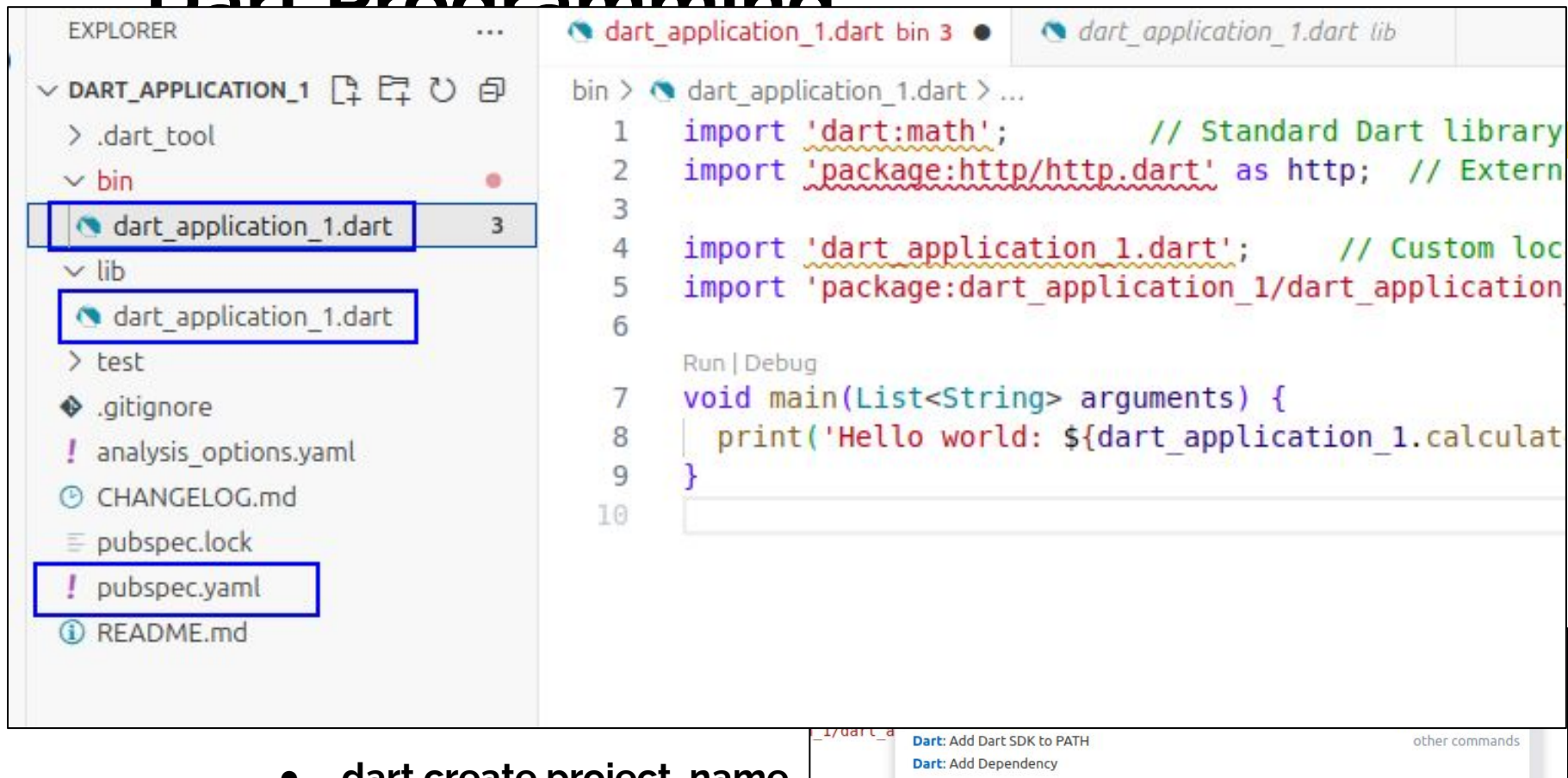
Dart: Add Dart SDK to PATH

Dart: Add Dependency

recently used ⚙

other commands

Dart Programming



The screenshot displays an IDE interface with two main panels. The left panel, titled 'EXPLORER', shows a project structure for 'DART_APPLICATION_1'. It includes a 'bin' directory containing 'dart_application_1.dart' (highlighted with a blue box and the number '3' in a grey box) and a 'lib' directory containing 'dart_application_1.dart' (highlighted with a blue box). Other files in the project include '.dart_tool', 'test', '.gitignore', 'analysis_options.yaml', 'CHANGELOG.md', 'pubspec.lock', 'pubspec.yaml' (highlighted with a blue box), and 'README.md'. The right panel shows the code editor for 'dart_application_1.dart'. The code is as follows:

```
bin > dart_application_1.dart > ...
1  import 'dart:math';           // Standard Dart library
2  import 'package:http/http.dart' as http; // Extern
3
4  import 'dart_application_1.dart'; // Custom loc
5  import 'package:dart_application_1/dart_application
6
Run | Debug
7  void main(List<String> arguments) {
8    print('Hello world: ${dart_application_1.calculat
9  }
10
```

At the bottom of the IDE, there are two tabs: 'dart_application_1.dart' and 'lib'. The 'lib' tab is active, showing the code for 'dart_application_1.dart'. The code includes imports for 'dart:math' and 'package:http/http.dart' as 'http'. It also includes a custom location import for 'dart_application_1.dart' and a package import for 'package:dart_application_1/dart_application'. The code defines a 'main' function that takes a list of arguments and prints 'Hello world: \${dart_application_1.calculat'.

- dart create project_name

Dart Programming Essentials

- Hello World in Dart

- Usually, you can hit the Run button to execute, but **prefer** to use the console with the command :

- **dart run OR dart bin/start_main.dart**

```
TERMINAL
▼ TERMINAL
o imed@imed-Inspiron:~/Dropbox/Workspace/Teaching/ENSIA/MobileDev/MyLectures/W6/hello_world_dart$ dart bin/start_main.dart
Type a number to compute the factorial :
5
Result for 5 ! is 120
Type a number to compute the factorial :
w
Error FormatException: Invalid radix-10 number (at character 1)
w
^
Type a number to compute the factorial :
```

Dart Programming Essentials

- **Dart Program Structure**

- Import Statements
- Variables
- Functions
- Classes
 - Variables
 - Constructors
 - Functions
- Entry Point : **main()** function

Dart Programming Essentials

- Dart Program Structure

- Import Statements

```
import 'dart:math';          // Standard Dart library as io, convert ..
import 'package:http/http.dart' as http; // External package that we add in pubspec.yaml

import 'dart_application_1.dart'; // Custom local file
import 'package:dart_application_1/dart_application_1.dart' as dart_application_1; // OR this way
Custom file

void main(List<String> arguments) {
  print('Hello world: ${dart_application_1.calculate()}!');
}
```

Dart Programming Essentials

- Dart Program Structure

- Import Statements

```
import 'dart:math';           // Standard Dart library as io, convert ..
import 'package:http/http.dart' as http; // External package that we add in pubspec.yaml

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}
```


Dart Programming Essentials

- **Dart Program Structure**

- Import Statements
- **Variables**
- Functions
- Classes
 - Variables
 - Constructors
 - Functions
- Entry Point : **main() function**

Dart Programming Essentials

- **Syntax of Dart : Variables**

- Declaring a variable can be set as
 - **Type name = value ;**
- **num** is to refer to both int and double
- **var** can be used to say **any** type
 - Can we assign a **var** to **int** and later **string** ?
- Const to create **immutable variables**
(unchangeable)
- Naming convention : **lowerCamelCase**

```
var myName = "Imed";  
String? sonName;  
String daughterName = "Amel";  
int age=2, size = 0, volume = 1; // Not recommended  
bool isHappy = true;  
const double pi = 3.14;  
num quantity = 2.1;
```

Dart Programming Essentials



The screenshot shows a code editor with the following Dart code:

```
1 void main() {  
2   var a=1;  
3   a="sss";  
4   print(a);  
5 }  
6
```

A blue "Run" button is visible next to the code. To the right, a "Console" window displays the following error message:

```
Error compiling to JavaScript:  
lib/main.dart:3:5:  
Error: A value of type 'String' can't be assigned to a variable of type  
    a="sss";  
      ^  
Error: Compilation failed.
```

- Can we assign a **var** to **int** and later **string** ?
 - Const to create **immutable variables**
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 - Naming convention : **lowerCamelCase**

Dart Programming Essentials

- **Syntax of Dart : Variables**

- Declaring a variable can be set as
 - **Type name = value ;**
- **num** is to refer to both int and double
- **var** can be used to say **any** type
 - Can we assign a **var** to **int** and later
- If you are forced to use dynamically typed variables (which is not recommended), you can use :
 - **dynamic a=1 ;**

```
var myName = "Imed";  
String? sonName;  
String daughterName = "Amel";  
int age=2, size = 0, volume = 1; // Not recommended  
bool isHappy = true;
```

```
1 void main() {  
2   dynamic a=1;  
3   a="sss";  
4   print(a);  
5 }  
6
```

Run

Console

sss

Dart Programming Essentials

- **Syntax of Dart : Variables**

- To check the type of a variable the keywords

- is
 - !is

```
void main() {  
  
  var f = 1;  
  if (f is int) {  
    print('f is an integer with value $f');  
  }  
  if (f is! String) {  
    print('f is not a string');  
  }  
}
```

Dart Programming Essentials

- **Syntax of Dart : Data Types and Converting**

- String
 - Use either single or double, but be consistent.
 - You can use the \$ sign inside a string for injecting variables into a string.
 - `print("Hello $name, my age is $age")`
 - You can use `"""` to create a multi-line string
- String to int : `int myAge=int.parse(strValue)`
- int to String : `String ageStr = myAge.toString()`
- String to double : `double myWeight=double.parse(strValue)`

Dart Programming Essentials

- **Syntax of Dart : Collections : List**

- List are like Arrays in Java defined as :
 - Growable List : `List<int> ages= [10, 21 ,31]`
 - Fixed Length List : `var ages=List<int>.filled(5,1) // Fills default val of 1`
- Accessing :
 - `ages[INDEX]`
 - Or even get inversely the index by using : `ages.indexOf(31)`
- Methods
 - **Ages.length** : to get the number of elements in the list



[]

Dart Programming Essentials

- **Syntax of Dart : Collections : List**

- Methods

- **ages.length** : to get the number of elements in the list
 - **add (item)** : to add an element
 - **insert(index, item)** : to add a specified index and shifting other elements
 - **remove(item)** : remove an item by value (only one item removed at a time)
 - **removeAt(index)** : remove element at a given index.

- Properties of the List:

- **first , last, isEmpty , isNotEmpty, reversed, single**

- Combining two list using **spread syntax**

- **var bigList =[...listOne, ...listTwo] ;**

[]

Dart Programming Essentials

- **Syntax of Dart : Collections : Set**

- Set is a unique collection of items. No duplicates (in contrast to a list)

```
Set<String> fruits = {"Apple", "Orange", "Mango"};
fruits.add("Grapes");
fruits.remove("Orange");
String firstElement = fruits.elementAt(0);
bool hasMango = fruits.contains("Mango");
int fruitsCount = fruits.length ;
```



{ }

Dart Programming Essentials



{key:val}

- Syntax of Dart : Collections : Map

```
Map<String, String> countryCapital = {  
    'USA': 'Nothing',  
    'India': 'New Delhi',  
    'China': 'Beijing'  
};  
  
countryCapital['Algeria'] = 'Algiers';  
bool checkOne = countryCapital.containsKey("India");  
bool checkTwo = countryCapital.containsValue("India");  
print("All keys of Map: ${countryCapital.keys}");  
print("All values of Map: ${countryCapital.values}");  
print("Is Map empty: ${countryCapital.isEmpty}");  
print("Is Map not empty: ${countryCapital.isNotEmpty}");  
print("Length of map is: ${countryCapital.length}");
```

Dart Programming Essentials

- **Syntax of Dart : Operations**

- Arithmetic Operators:
 - % Modulus
 - ~/ Integer division (Divide two numbers and give output as int)
- Logical Operators
 - && and
 - || or
 - ! not
- Type Test Operators
 - is (myValue is **int**)
 - is! (myValue **is!** String)

Dart Programming Essentials

- Syntax of Dart : Functions

```
//function to add
int add(int a, int b) {
    var total;
    total = a + b;
    return total;
}
```

Dart Programming Essentials

- Syntax of Dart : Functions

- Arrow Functions

(For SINGLE LINE expressions)

```
int square(int x) => x * x;
```

```
int cube(int x) {  
    return x * x * x;  
}
```

Dart Programming Essentials

- Syntax of Dart : Functions
 - With Named Parameters

```
void greet({required String name, int age = 20}) {  
  print('Hello, $name. You are $age years old.');
```



```
void main(List<String> arguments) {  
  greet(name: 'Sara', age: 25); // Must provide both  
}
```

Dart Programming Essentials

- Syntax of Dart : Functions
 - Anonymous Functions (Lambdas)

```
void main(List<String> arguments) {  
    var numbers = [1, 2, 3];  
    numbers.forEach( (n) {  
        print(n * n);  
    });  
}
```

Dart Programming Essentials

- Syntax of Dart : Function

- Callbacks
(Function as a
Parameter
to another
function)

```
void execute(Function func) {  
    func(); // Call the function  
}  
  
void sayHello() => print('Hello!');  
  
void main() {  
    execute(sayHello); // Works fine  
}
```


Dart Programming Essentials

- Syntax of Dart : Functions

- Callbacks

(Function as a
Parameter
to another
function)

```
int multiply(int a, int b) {  
    return a * b;  
}  
  
// A function that takes another function as a parameter  
void calculate(int x, int y, int Function(int, int) operation)  
{  
    print('Result: ${operation(x, y)}');  
}  
  
void main() {  
    calculate(4, 5, multiply); // Output: Result: 20  
}
```

Dart Programming Essentials

- Syntax of Dart : Flow Control : IF-ELSE + SWITCH

```
var noOfMonth=1;
if (noOfMonth == 1) {
    print("The month is jan");
} else if (noOfMonth == 2) {
    print("The month is feb");
} else {
    print("Invalid option given.");
}
```

```
var dayOfWeek = 5;
switch (dayOfWeek) {
    case 1:{
        print("Day is Sunday.");
        break;
    }
    case 2:{
        print("Day is Monday.");
        break;
    }
    Default:{
        print("Invalid Weekday.");
        break;
    }
}
```

Dart Programming Essentials

- Syntax of Dart : Flow Control : For + While

```
for (int i = 1; i <= 10; i++) {  
    print(i);  
}
```

```
List<int> numbers = [1,2,3,4,5];  
int total = 0;  
numbers.forEach( (num)=>total= total+ num);
```

```
List<int> numbers = [1,2,3,4,5];  
for (int a in numbers){  
    print(a);  
}
```

```
int i = 1 ;  
while (i <= 10) {  
    print(i);  
    i=i+1 ;  
}
```

Dart Language

- Syntax of Dart :
Object Oriented
Programming

```
class Person {  
    String name;  
    int age;  
  
    // Constructor  
    Person(this.name, this.age);  
  
    // Method  
    void greet() {  
        print('Hello, my name is $name and I am $age years old.');    }  
}  
  
void main() {  
    var person = Person('Imed', 30);  
    person.greet(); // Output: Hello, my name is Imed and I am 30 years old.  
}
```

Dart Language

- Syntax of Dart :
Object Oriented
Programming

```
class Person {  
    String name;  
    int age?;  
  
    // Constructor  
    Person(this.name, this.age);  
    // Method  
    void greet() {  
        print('Hello, my name is $name and I am $age years old.');    }  
}  
  
void main() {  
    var person = Person('Imed', null);  
    person.greet(); // Output: Hello, my name is Imed and I am null years old.  
}
```

Dart Language

- **Syntax of Dart :**

. : Dot Operator Used to
invoke a function on an
Object or access its attribute

.. : Cascade operator :
Same as the Dot , but stay
At the same level (Object)

..? : Proceed only if the object is not null.

```
var person = Person();  
person.name = 'Imed';  
person.age = 30;  
person.greet();
```

```
var person = Person()  
  ..name = 'Imed'  
  ..age = 30  
  ..greet();
```

Dart Language

- Inheritance

```
class Animal {  
  void speak() => print('Animal makes a sound');  
}  
  
class Dog extends Animal {  
  @override  
  void speak() => print('Barking..!');  
}  
  
void main() {  
  Animal dog = Dog(); //or var dog = Dog();  
  dog.speak(); // Barking..!  
}
```

Dart Language

- Setters & Getters

```
class Rectangle {  
    double width;  
    double height;  
  
    Rectangle(this.width, this.height);  
}  
  
void main() {  
    var rect = Rectangle(4, 5);  
    rect.width = 10;    // directly change attribute  
    print(rect.width); // 10  
}
```


Dart Language

- Setters & Getters

```
class Rectangle {  
  double width, height;  
  Rectangle(this.width, this.height);  
  // Getter  
  double get area => width * height;  
  // Setter  
  set changeWidth(double newWidth) {  
    width = newWidth;  
  }  
}  
  
void main() {  
  var rect = Rectangle(4, 5);  
  print('Area: ${rect.area}'); // 20  
  rect.changeWidth = 10;  
  print('New Area: ${rect.area}'); // 50  
}
```

Dart Programming Essentials

- Syntax of Dart : Exception Handling

```
try {  
    int number = int.parse(line);  
    var result = computeFactorial(number);  
    print("Result for $number ! is $result");  
} catch (e, stacktrace) {  
    print("Error ${e.toString()} stackTrace $stacktrace ");  
}
```

Dart Programming Essentials

- Syntax of Dart : Reading from the Console.

```
import 'dart:io';

void main() {
  // Ask for user input
  stdout.write('Enter your name: ');
  // Read input from console
  String? name = stdin.readLineSync();
  // Print the result
  print('Hello, $name!');
}
```



References

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